

Subtracting Two-Digit Numbers with Place-Value Blocks

Answer Key

Grades 2–3

Quick Answers

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| 1. the difference = 24 | = 33 |
| 2. the difference = 33 | 6. the marbles left = 37 marbles |
| 3. the difference = 25 | 7. (a) the correct difference = 36, — |
| 4. the difference = 36 | 8. (a) the estimate = 40, (b) the exact difference |
| 5. (a) the estimate = 30, (b) the exact difference | = 47, — |
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What is verified

6 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 2–3
2.NBT.B.5 – problems 1, 2, 3, 4, 5, 6, 7, 8
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

- 3. *If they got 35: took the smaller digit from the larger in each column instead of regrouping*
 - 4. *If they got 44: took the smaller digit from the larger in each column instead of regrouping*
 - 5. *If they got 40: rounded both numbers down instead of to the nearest ten*
 - 7. *If they got 44: took the smaller digit from the larger in each column instead of regrouping*
 - 8. *If they got 50: rounded the number being taken away down instead of up*
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1. The blocks show 47. Cross out 23. What is left?

Solution: There are enough ones to take from, so no rod needs opening. Cross out three of the seven single squares, leaving four ones, then cross out two of the four rods, leaving two rods. Two tens and four ones.

24

Blocks left: two rods and four squares.

2. The blocks show 68. Cross out 35. What is left?

Solution: Ones first: cross out five of the eight squares, leaving three. Then three of the six rods, leaving three rods. Three tens and three ones. 33

3. The blocks show 52. Cross out 27.

Solution: There are only two single squares and seven must go, so open one rod into ten ones. The picture now shows four rods and twelve ones — still fifty-two, only shown differently. Cross out seven of the twelve ones, leaving five; cross out two of the four rods, leaving two. Two tens and five ones. 25

4. The blocks show 64. Cross out 28.

Solution: Four ones cannot give up eight, so open a rod: five rods and fourteen ones. Take eight ones, leaving six; take two rods, leaving three. Three tens and six ones. 36

5. The blocks show 71. (a) Estimate $71 - 38$ to the nearest ten. (b) Find the exact answer.

Solution (a): 71 rounds down to 70 and 38 rounds up to 40. Subtracting the rounded numbers gives $70 - 40$. 30

Solution (b): One single square cannot give up eight, so open a rod: six rods and eleven ones. Take eight ones, leaving three; take three rods, leaving three. The exact answer sits three above the estimate, so the estimate did its job. 33

6. A jar held 82 marbles and 45 were taken out. How many are left?

Solution: “Taken out” means subtract. Two ones cannot give up five, so open a rod: seven rods and twelve ones. Take five ones, leaving seven; take four rods, leaving three. 37 marbles

7. Sam worked out $63 - 27$ and wrote 44. (a) Work it out correctly. (b) Explain what Sam did wrong.

Solution (a): Three ones cannot give up seven, so open a rod: five rods and thirteen ones. Take seven ones, leaving six; take two rods, leaving three. 36

Solution (b) — instructor-judged. A correct response names the flipped column. Full credit for the idea that there were only three ones to take seven from, and instead of opening a rod the ones column was worked backwards — the smaller digit taken from the larger. Accept an answer that says a ten should have been opened into ten ones first, leaving five rods and thirteen ones. Saying only that the answer is too big, or that regrouping was needed, without naming the reversed column, earns half credit. Restating the correct difference earns no credit, since part (a) already asks for it.

8. The blocks show 94. (a) Estimate $94 - 47$ to the nearest ten. (b) Find the exact answer. (c) Explain why the estimate landed below it.

Solution (a): 94 rounds down to 90 and 47 rounds up to 50, so the estimate is $90 - 50$. 40

Solution (b): Four ones cannot give up seven, so open a rod: eight rods and fourteen ones. Take seven ones, leaving seven; take four rods, leaving four. 47

Solution (c) — instructor-judged. A correct response tracks the direction of each rounding. Full credit for the idea that the number being taken away was rounded *up* while the number in front was rounded *down*, so the estimate takes away more from less and must come out low. Accept an answer that follows just one of the two roundings correctly and says which way it moved the result. Noticing only that the estimate is smaller, with no

reason, earns half credit. Claiming that an estimate must always match the exact answer earns no credit.